Computers and Electrical Engineering 104 (2022) 108393

Contents lists available at ScienceDirect

Computers and Electrical Engineering

journal homepage: www.elsevier.com/locate/compeleceng

Intelligent overall planning model of underground space based on digital twin

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ARTICLE INFO

Keywords:
Digital Twin Technology
Overall Planning of Underground Space
Construction Settlement Law
Disaster Prevention Monitoring System
Intelligent Planning

ABSTRACT

Against the background of the rapid development of social economy and advanced industrial manufacturing, emerging technologies are sweeping the world. The overall planning of urban space is also gradually developing in the direction of "intelligence". The real estate market has been booming in recent years. As a result, the available land resources on the surface are declining. It first gives a general introduction to the digital twin technology and operating rules, and then introduces the analytic hierarchy process and the complexity evolution process model for specific analysis. It first determines the relative weight vector, and then verifies the consistency of the judgment matrix to obtain the simulated probability density function. For the calculation of underground space safety accident risk assessment, the Bayesian network is used for evaluation. This facilitates disaster warnings of possible and ongoing accidents and reduces the adverse consequences of accidents. Then, the digital twin technology is applied to the analysis of underground space visualization, underground construction and underground disaster prevention monitoring. During the construction stage, the bottom of the upper structure column has changed a lot. The difference between the difference between the foundation difference between the adjacent color spectrum column of the upper structure is relatively mild. When digging a large area of ground soil, the foundation dropped to 5.6 mm. When the soil on the side of the pile was removed, the ground building was basically settle down, 0.8 mm. When the structure of the pile side soil is completed, it indicates that the technology of the pile foundation base of the surface building is basically. Finish. Subsequent sidewalk soil has almost no effect on the upper structure. At this stage, the final deposition difference is 2.427 mm, which meets the requirements of the neighboring column base in the current regulations. Finally, I studied the prevention of disaster prevention in the underground space, simulated the accident scene, and obtained the most suitable imaging equipment for the monitoring of disaster. After the prevention of disaster testing and simulation accidents for underground space, the most suitable imaging equipment for preventing disaster testing was obtained. Its formal digital twin technology.

1. Introduction

New technologies represented by big data, cloud computing, and artificial intelligence are developing rapidly. Driven by national and government industrial policies, new technologies are rapidly infiltrating into multiple fields and industries. While the new

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https://doi.org/10.1016/j.compeleceng.2022.108393
Received 27 May 2022; Received in revised form 6 September 2022; Accepted 9 September 2022
Available online 18 October 2022
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technology and the real economy are deeply integrated, the underground urban space is also gradually developing towards "intelligence". The underground space mainly includes the utilization of underground pipeline construction, subway construction, underground commercial street, parking lot, and underground civil air defense facilities. The underground space is relatively vast and has broad prospects, and the demand for development and utilization is also relatively strong. However, there are not many studies on the intelligent overall planning of underground space. In addition, the construction of underground space is difficult, the cycle is long, the cost is high, and the comprehensive social impact is large, so that a systematic and complete overall management method has not yet been formed. At the same time, with the development of society and the development of modernization, the population has exploded. The surface area obviously cannot meet the needs of social construction. Therefore, it is an important measure to meet the social development to carry out the comprehensive development and management strategy research of underground space from all aspects.

In recent years, the real estate market, especially in large and medium-sized cities, has been very hot. This has led to a sharp decrease in surface land resources, and the coordinated development of underground spaces has become more and more common. At present, most cities and towns have incorporated the planning and construction of underground space into the long-term development strategy of the region. This has greatly promoted the development of the region. Therefore, accelerating the process of intelligent overall planning of underground space not only has certain theoretical value for social development, but also has practical guiding significance for surface land use. This realizes the digitization and virtualization of elements in the space, the real-time visualization of all states, the coordination and intelligence of management decisions, and forms a development pattern in which the physical world and the virtual world coexist, and the virtual and reality blend. Digital twin technology has previously been used more in the study of above-ground space, and less attention has been paid to underground space. But in fact, underground space is a very important natural resource and an important component of urban space. Therefore, accelerating the intelligent overall planning of underground spaces based on digital twins has far-reaching significance for the development of society.

This paper uses digital twin technology to carry out a comprehensive planning of underground space, build a systematic architecture and clarify the planning process.

The innovation of this paper is that (1) It studies the model of intelligent overall planning of underground space based on digital twin technology. (2) The five -dimensional model of the twins in digital twin technology to build a visual discrimination model for underground space monitoring and management, which provides a strong guarantee for managers' management and decision -making (3). It proposes the analytic hierarchy process and the calculation model of underground space safety accident risk assessment, and uses the Bayesian network to evaluate the risk accident. This is more effective than other methods in saving cognitive time and increasing coping speed.

2. Related work

At present, the rapid development of science and technology has promoted the progress of space technology, and more and more people have studied it. Among them, Moridi M A used wireless sensor network (WSN) to analyze and study tunnel and underground space technology. The authors introduce static node deployment of ZigBee topology in 12, 20, 30, 40 and 50 node scenarios for performance analysis and performance evaluation in underground environment. It includes throughput, packet delivery rate (PDR), end-to-end latency, energy consumption, and packet delivery security. However, this method requires higher operation in practice [1]. Ding Z studied a new type of mobile flood wall. In the detection of a subway station, the model test was used to verify the safety and anti-leakage capability of the model, and the safety factor $k = 1.14>1$ was obtained, which met the strength requirements. However, this process is complicated and requires a large amount of accurate data as support [2]. Lv X reveals the key factors affecting the construction of underground spaces and establishes theoretical relationships between torque, propulsion and soil strength parameters (i.e. unconfined compressive strength, shear strength and cohesion). It uses 3D flexible boundary loading device to carry out 75 rotary excavation model laboratory tests on 15 groups of silty clay. According to the test results, the theoretical relationship was simplified, the model parameters were determined, and 15 quantitative relationships with 5 different water contents (4.8%) were obtained. Finally, the quantitative correlation was verified by field application. The results show that it is feasible to predict soil in-service strength [3]. Yamaguchi K simulated three-dimensional groundwater flow around an abandoned mine to assess the impact of backfilling the mine's underground space on reducing acid mine drainage (AMD). The authors calculated steady-state groundwater flow for two days of high and low rainfall to calibrate the model. The calculated AMD groundwater level and flow rate are consistent with the measured results. Results when backfilling the excavated underground space showed that AMD's flow was reduced by 5% to 30%. This suggests that backfilling the excavation space is one of the effective ways to reduce the amount of discarded AMD. However, this evaluation model has certain errors and the accuracy is not high enough [4]. Wang Z's study used the CRD (Cross Diagram) method and the four-step method to excavate large-diameter shallow-buried double tunnels in soft soil foundations. According to the on-site monitoring results, the effects of construction factors such as tunnel excavation method and construction speed, and environmental factors such as soil conditions and continuous rainfall on the settlement of the ground surface and tunnel vault are analyzed. For construction factors, it is shown that the surface subsidence caused by the four-step method is 332% of that caused by the argillaceous siltstone CRD method. However, due to the complex environmental factors on site, the requirements for the machine are relatively high [5]. Peng J B studied the three-dimensional motion generated by ground fissures in Xi'an, among which vertical displacement has the greatest impact on subway engineering. The results of large-scale physical model tests show that the activity of ground fissures may lead to damages such as tunnel lining cracks, waterproof failure and subway track deformation. In order to alleviate such risks, the author proposed a series of engineering countermeasures, such as expanding the tunnel section to maintain clearance, adopting segmented tunnels, strengthening tunnel linings, and adopting various waterproof measures [6]. Lee Y K used the factor of safety (FOS) index to quantify the degree of safety of various rock structures. The authors used the Mohr-Coulomb and Hoek-Brown failure criteria

to define four local safety factors based on maximum shear stress, maximum shear strength, stress invariants, and maximum principal stress. Then, the variation characteristics of each FOS along five stress paths were investigated. The results show that local FOS based on maximum shear stress or stress invariant is more necessary than local FOS based on shear strength when conservative rock mechanics design is required. This has practical significance for ensuring the safety of underground space. However, because this method is relatively old, the data obtained is slightly lacking, and the necessary innovation is lacking [7].The study of the above scholars played a pivotal role in the overall planning project of the subsequent underground space.However, there are many fields of the above research in the underground space, but it has not been mentioned in terms of intelligent planning.

3. Overall planning method of underground space based on digital twin

3.1. Digital twin model

(1) Five elements of the digital twin model

Digital twin is also called digital twin and digital mapping. It is a large number of physical and digital models correspondingly generated by enterprises in the process of implementing model-based systems engineering [8]. As the mapping of virtual space to physical products, it reflects the whole process of introduction, growth, maturity and decline of physical products. Digital twin consists of five elements: service system, physical entity, virtual model, information connection, and twin data. It evolves through mutual learning between related digital models. It also improves the decision making of the winners during the life cycle of the physical entity [9,10]. Fig. 1 shows the five-dimensional model of the digital twin.

When working with digital twins, the physical entity is first analyzed in terms of requirements and application objects. Then, the corresponding virtual digital twin model is constructed, and the interactive connection is established to realize the data interaction between the virtual and the real and the service system [11]. It uses big data to analyze twin data, and finally provides users with the required service system. The detailed composition of the five elements of the digital twin is shown in Fig. 2.

Digital twins should follow the following rules. The "physical entity" is the application carrier. The entire model should be based on physical entities and run with them. "Virtual model" is the key core to ensure the realization of model functions. "Interconnection" is a topological network that connects various elements into a whole, and connects various elements to realize information exchange. The "twin data" is the key element that drives the operation of the entire model. "Service system" is a data analysis tool, responsible for data calculation and model improvement. It helps users to judge and make decisions.The above elements are derived from the framework of the digital twin technical architecture. In terms of digital twin fields, the five elements have realized the functional module of digital twin technology.

(2) Construction criteria of digital twin model

The digital twin model has the construction criteria of "four modernizations, four possibilities, and eight uses" [12]. In the process of operation, it should meet the requirements of "four possibilities" and "four modernizations". The final result should meet the standard of "eight uses". The digital twin model construction guidelines are shown in Fig. 3.

Digital twins use sensors to perceive real-time changes in physical entities by constructing virtual models that map physical entities and their inherent laws [13]. The digital twin model can predict the changing trends of physical entities and allow the physical entity space to perceive these changes in advance. When making decisions in the physical entity space, the optimal control scheme can be selected in advance for the physical entity. Therefore, it is very meaningful to apply digital twin technology to the intelligent overall planning of underground space.

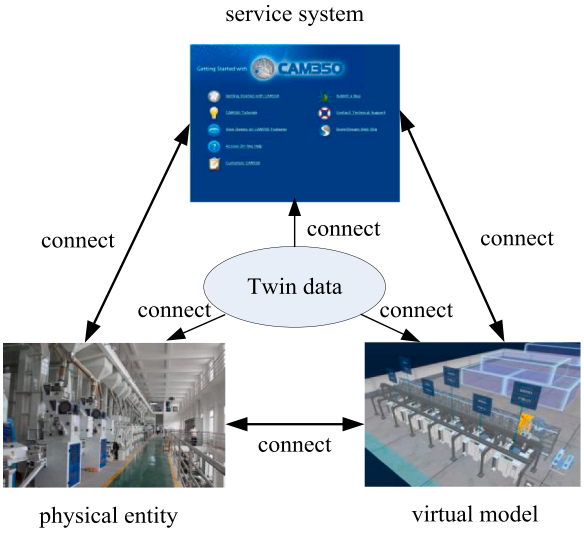


Fig. 1. Digital twin five-dimensional model.

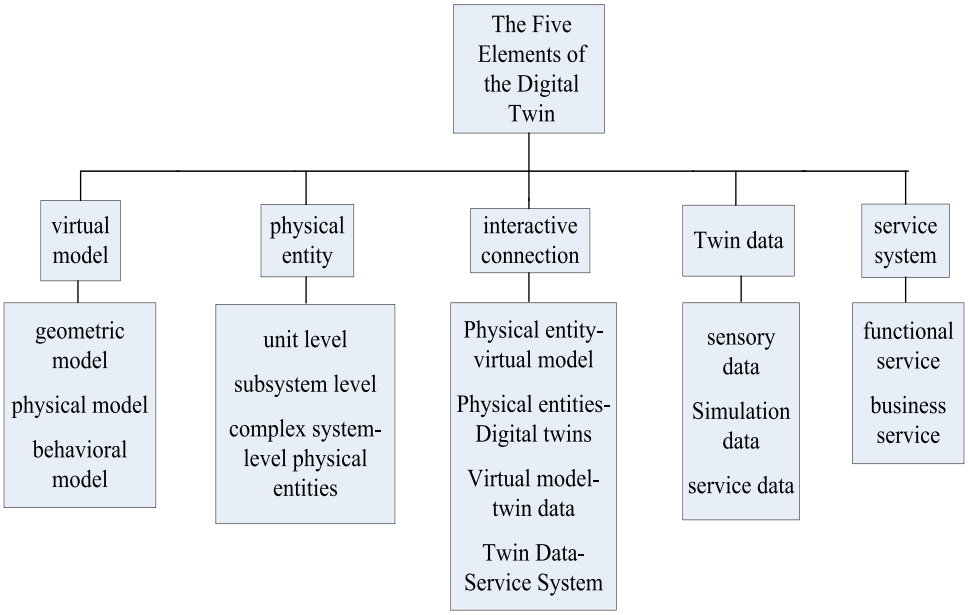


Fig. 2. Detailed composition of five elements of digital twin.

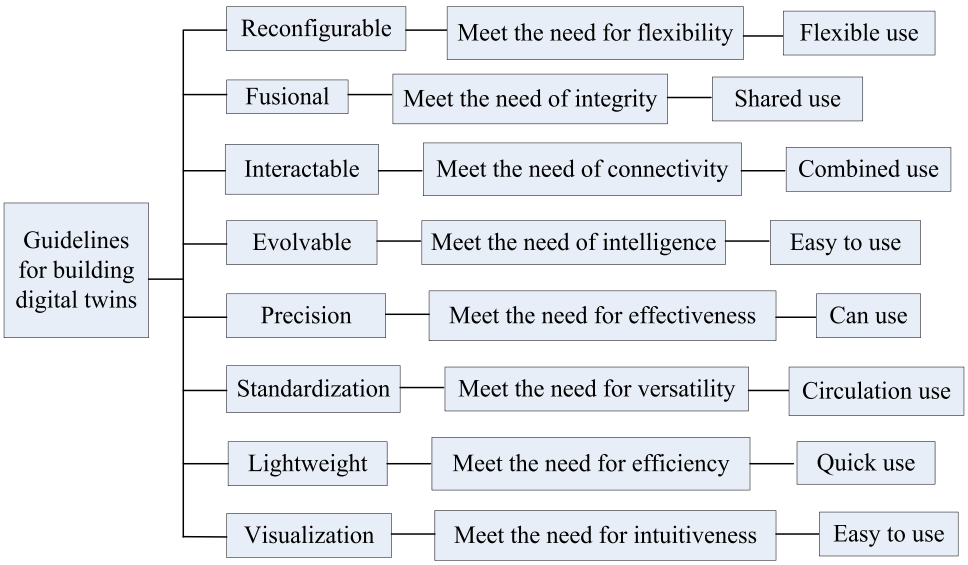


Fig. 3. Guidelines for building a digital twin model.

3.2. Analytic hierarchy process

Affected by complex geological factors and underground structures, the development of underground space has certain difficulties. Geological factors have layers, ambiguities and differences [14]. Because the construction cost of underground projects has considered many factors, the cost is about 4 times the ground project . The intelligent overall planning model of underground space needs to be able to accurately reflect these characteristics. Analytic Hierarchy Process (AHP) can better analyze such complex factors. It stratifies complex problems, and finally selects the optimal solution.

First, it is assumed that the proportion of n factors x_1,x_2,..., x_n in the upper layer C should be determined, and their importance to C should be compared. Assuming $x_{jk} = x_j/x_k$, therefore, a judgment matrix $x_{jk} = x_j/x_k$ is obtained. As shown in Table 1.

It can be seen that $x_{jk} > 0$, $x_{jk} = 1/x_{jk}$, $x_{jk} = 1/x_{jk}$, $x_{jj} = 1$ (j,k = 1,2…,n).

(1) Relative weight vector determination

It uses the sum-product method. It takes the normalized arithmetic mean of the n column vectors of the judgment matrix as the

weight, and obtains the weight vector Q. That is

$$Q_j = \frac{1}{n}\sum_{k=1}^{n}\frac{x_{jk}}{\sum_{j=1}^{n} x_{jk}} (j = 1, 2, ..., n) \tag{1}$$

For the general judgment matrix U, it have

$$U \cdot Q = \lambda_{\max} Q \tag{2}$$

$$\lambda_{\max} = \sum_{j=1}^{n}\frac{(U \cdot Q)_j}{nq_j} = \frac{1}{n}\sum_{j=1}^{n}\frac{\sum_{j=1}^{n}\alpha_{jk}q_k}{q_j} \tag{3}$$

Among them, $(U \cdot Q)_j$ represents the jth component of $U \cdot Q$, $\lambda_{\max}$ is the largest eigenroot of U, and Q is the eigenvector of $\lambda_{\max}$.

(2) Consistency test

In the actual operation of the model, it is necessary to continuously verify the consistency of the judgment matrix until the judgment matrix meets the requirements. The consistency index of the judgment matrix is:

$$XA = \frac{\lambda_{\max} - 1}{n - 1} \tag{4}$$

The smaller the value of XA, the less the judgment matrix deviates from perfect consistency. When XA=0, the judgment matrix is completely consistent. N is the order, and the smaller n is, the smaller the value of XA is. The random consistency index VA is obtained through practical experience, as shown in Table 2.

Consistency ratio metrics:

$$XV = \frac{XA}{VA} \tag{5}$$

When XV<0.01, the consistency meets the requirements. Otherwise, readjustment is required until XV < 0.01.

(3) Combined weight vector

Assuming that the weights of n factors in the middle layer X layer to the total target layer C are ranked as x_1,x_2,..., x_n, and the m factors in the L layer are ranked as l_{1k},l_{2k},..., l_{mk}($k = 1, 2, ..., n$) in the hierarchical order of the factor x_k in the upper layer C, then the weight of the jth factor in the L layer to the total target is $\sum_{k=1}^{m} x_k l_{jk}$, as shown in Table 3.

Assuming that the single-level ordering consistency index of L layer l_1,l_2,..., l_m to element X_k($k = 1, 2, ..., n$) in the upper layer is XA_k, and the random consistency index is XA_k, then the consistency ratio of the total ordering of the hierarchy is:

$$XV = \frac{x_1 XA_1 + x_2 XA_2 + ... + x_n XA_n}{x_1 VA_1 + x_2 VA_2 + ... + x_n VA_n} \tag{6}$$

When XV < 0.1, the consistency meets the requirements, otherwise it is necessary to readjust the value with a higher ratio.

(4) Variable weight model

After the weight of each factor is obtained by the upper-level analysis method, the overall planning of the underground space is usually evaluated according to the weighted sum model:

$$p = \sum_{j=1}^{n} q_j s_j (j = 1, 2, ..., n) \tag{7}$$

Among them, P is the comprehensive evaluation score, and q_j is the combined weight when the index is j. s_j is the normalized value when the index is j.

But this model has certain errors. Better indicators in the evaluation unit will neutralize poor indicators, making the overall result better. In the actual overall planning, when other indicators are particularly good, once one indicator is particularly poor, it cannot be considered as being suitable [15]. Therefore, in order to make the planning results more reasonable, this paper introduces the idea of weight.

Assuming that the vector $G^0 = (G_1^0, G_2^0, ..., G_u^0)$, such as $G_j^0 \in (0,1)$, satisfies $\sum_{j=1}^{u} G_j^0 = 1$, then G^0 is called a u-dimensional constant

Table 1
Comparison table of judgment matrix units.

C	x_1	x_2	...	x_k	...	x_n
x_1	x_{11}	x_{12}	...	...	...	x_{1n}
x_2	x_{21}	x_{22}	...	...	...	x_{2n}
...	...	...	...	...	...	...
x_j	...	...	...	x_{jk}	...	x_{jn}
...	...	...	...	...	...	...
x_n	x_{n1}	x_{n2}	...	x_{nk}	...	x_{nn}

Table 2
Average random consistency index.

n	1	2	3	4	5	6
VA	0	0.12	0.67	1.25	1.38	1.46
n	7	8	9	10	11	12
VA	1.54	1.58	1.61	1.63	1.65	1.66

Table 3
Calculation of the weights of the L layer to the C layer.

X L	$x_1,x_2,\ldots, x_n$ $x_1,x_2,\ldots, x_n$	Level of L layer Total sorting
l_1	$l_{11}l_{12}\ldots l_{1n}$	$\sum_{k=1}^{n} x_k l_{1k} = l_1$
l_2	$l_{21}l_{22}\ldots l_{2n}$	$\sum_{k=1}^{n} x_k l_{2k} = l_2$
…	…	…
l_m	$l_{m1}l_{m2}\ldots l_{mm}$	$\sum_{k=1}^{n} x_k l_{mk} = l_m$

weight vector. A set of dimensionally variable weights refers to u mappings u_j:

$$G_k : [0,1]^m \to (0,1), (i_1, i_2, \ldots, i_u)| \to g_k(i_1, i_2, \ldots, i_u) \tag{8}$$

If there is T: $[0, 1]^u \to (0, +\infty)^u$,$I \to T(I) = (T_1(I),T_2(I),\ldots, T_u(I))$, such as:

$$T_1 : i_j \geq i_k \Rightarrow T_j(I) \leq T_k(I); \tag{9}$$

$$T_{1'} : i_j \geq i_k \Rightarrow T_j(I) \geq T_k(I); \tag{10}$$

$$T_{1''} : 1 \leq j,k \leq r, i_j \geq i_k, T_j(I) \leq T_k(I)\ r+1 \leq j,k \leq u, i_j \geq i_k, T_j(I) \geq T_k(I) \tag{11}$$

T_2: $T_k(I)$ is continuous for each argument i_k;
T_3: For any constant weight vector $G_1^0 = g_1^0, g_2^0, \ldots, g_u^0$, it can get:

$$G(I) = \frac{\left(g_1^0 \cdot T_1(I), g_2^0 \cdot T_2(I), \ldots, g_u^0 \cdot T_u(I)\right)}{\sum_{k=1}^{u}(g_k^0 \cdot T_k(I))} = \frac{G^0 \cdot T(I)}{\sum_{k=1}^{u}(g_k^0 \cdot T_k(I))} \tag{12}$$

T(I) is an m-dimensional state variable weight vector. G^0 is the indicator constant weight vector. G(I) is the normalized product of the factor constant weight vector G^0 and the state variable weight vector T(I).

If T_1,T_2,T_3 is satisfied, then $T(I)$ is a penalized state variable weight vector. If $T_{1'}$,T_2,T_3 is satisfied, then $T(I)$ is an excitation-type state variable weight vector. If $T_{1''}$,T_2,T_3 is satisfied, then $T(I)$ is a mixed state variable weight vector;

3.3. Complexity evolution process model

The simulated probability density function can be obtained from the real-time detection data of the digital twin in the underground space. There are many uncertainties in the overall planning of underground space. This accumulating uncertainty can cause serious problems for development and construction [16]. After the probability density function is calculated, it can be calculated according to the probability solution formula that changes with time. The formula is:

$$F_t^v = \int c^t(WS)dWS \tag{13}$$

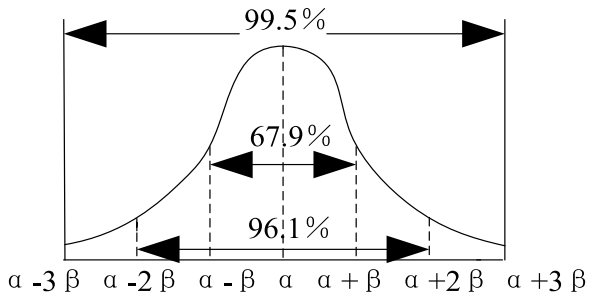


Fig. 4. Normal distribution diagram.

(1) Normal distribution

The random variables in most experiments are normally distributed. The mean of the sum of different random variables also exhibits a normal distribution [17]. Therefore, the normal distribution of the probability density function of the physical quantity parameters related to the functional elements of the system is calculated as shown in Fig. 4.

From Fig. 5, when the measuringtor reaches 100 rpm, the probability density reaches 0.08, while the density of 90 rpm/minute and 110 rpm, density, density But 0.

The probability power function of the normal distribution is:

$$f(x) = \frac{1}{\sqrt{2\pi\beta}} \exp\left(- \frac{(i-\alpha)^2}{2\beta^2} \right) \tag{14}$$

Among them, α is the mean of a normally distributed random variable and β is the standard deviation of the random variable.

After it is determined that the physical quantity parameters corresponding to the functional elements belong to the normal distribution, the data detected by the digital twin model is used for data analysis and calculation. For example, the rotational speed of the underground surveyor is the output value of the physical quantity parameter corresponding to the functional element. It needs to be calculated according to the physical quantity parameters input by the surveyor. The rotating speed design range of the surveyor is 150 ±5 r/min. By fitting these data with MATLAB, the probability density function of the surveyor can be obtained, as shown in Fig. 5.

(2) Uniform distribution

Within each range, the physical quantity parameter values are equal. That is, the probability of the physical quantity parameter appearing in each interval is equal. The probability density function for a uniform distribution is:

$$f(x) = \left\{ \begin{array}{l} \frac{1}{\beta - \alpha}, \alpha \le x \le \beta \\ 0, x < \alpha, x > \beta \end{array} \right\} \tag{15}$$

After the probability density function of the functional element and the system scope of the functional element are determined, the function realization rate of the surveyor can be obtained. Using the digital twin data to fit a probability density function, the deviation of the functional system range compared to the design can be determined.

(3) Risk assessment and calculation of underground space safety accidents

The occurrence of a risk accident means that the risk factor has become a reality [18]. Consequences such as casualties, property damage, adverse social impacts, etc., may even cause damage to surface buildings. Using Bayesian network for risk accident evaluation is beneficial to accurately control the adverse consequences caused by risk accidents.

1) The basic formula of Bayesian network

$0 \le g(\mathrm{X}) \le 1$; G(Ture) = 1; G(False) = 0; (16)

Addition formula:

$$\mathrm{G}(X+Y) = G(\mathrm{X}) + G(\mathrm{Y}) - G(\mathrm{X} \cap \mathrm{Y}) \tag{17}$$

X,Y(X∩Y) means that event X occurs and event Y occurs. $X + Y$ (X∪Y) means that event X occurs or event Y occurs.

Conditional probability:

$$G(X|Y) = \frac{G(X,Y)}{G(Y)} \tag{18}$$

Formula 18 represents the probability of occurrence of event X assuming that another event Y has already occurred.

Conditional independence:

$$\text{if } \mathrm{X} \perp \mathrm{Y}(\text{Independence}),\ \mathrm{G}(\mathrm{X}|\mathrm{Y}) = G(\mathrm{X}) \tag{19}$$

Joint probability:

G(x, y) = G(x|y)G(y)

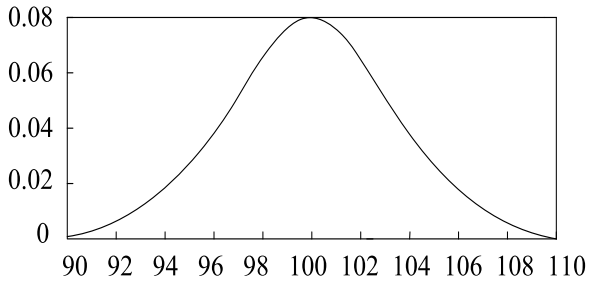


Fig. 5. Normal distribution of the surveyor rotational speed.

G(X, Y, Z) = G(x, P) = G(x|P)G(P) = G(x|y, z)G(y, z)
= G(x|y, z)G(y, z)G(z) (20)

2) Algebraic expression of Bayesian network

The joint probability formula of n variables is extended by formula (18), formula (19) and formula (20):

$$G(i_1, i_2, ..., i_n) = \prod_{k=1}^{n} G(i_k|parents(i_k)) \tag{21}$$

Among them, *parents*(i_k) represents all ancestor elements of element i_K, or can be represented as:

$$G(I_1 = i_1, ..., I_n = i_n) = \prod_{k=1}^{n} G(I_k = i_k|I_{k+1} = i_{k+1}, ..., I_n = i_n) \tag{22}$$

Combining the above, the joint distribution of x, y, z, q, r, s, t can be obtained:

$$G(x)G(y|x)G(z|x, y)G(q|x, y, z)G(r|x, y, z, q)G(s|x, y, z, q, r)G(t|x, y, z, q, r, s)) \tag{23}$$

The joint distribution simplifies G(x)G(y)G(z)G(q|x, y)G(r|x, z)G(s|q)G(t|q, r).

The joint distribution of all variables is obtained from $G(I_1 = i_1, ..., I_n = i_n) = \prod_{k=1}^{n} I_k = i_k|I_{k+1} = i_{k+1}, ...I_n = i_n)$:

$$\begin{aligned} &G(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13 \\ &= \prod_{k=1}^{13} G(I_k = i_k|I_{k+1} = i_{k+1}, ..., I_n = i_n) \\ &= G(1)G(2)G(3)G(4)G(5|3, 2, 1)G(6)G(7|4)G(8|7, 2, 1)G(9|8, 7, 6) \\ &G(10|8, 5, 3, 2, 1)G(11|10, 8, 7, 5)G(12|11, 10, 9, 7)G(13|11, 10, 9, 8, 7, 5) \end{aligned} \tag{24}$$

It can be seen from the above algorithm that hierarchical analysis helps analyze the comprehensive humidity time data table and temperature change analysis of the underground space. The consistency test method helps the study of the bottom of the column.

4. Experiment of intelligent overall planning of underground space

4.1. Visualization of underground space

(1) Time data table

Time-based data tables are an important way to visualize and manage underground spaces. The data sheet is mainly responsible for statistics and summary analysis in underground space safety management. It is helpful for managers to grasp the situation of underground space changing with time [19]. Under the safety monitoring of underground space, the data table can summarize the data changes of key targets over a period of time. For example, for the monitoring of the integrated pipe gallery in the underground space, the time data table can make statistics on the temperature and humidity of the integrated pipe gallery based on time. This makes it easy for managers to control the temperature and humidity of the integrated pipe gallery over time and solve problems. As shown in Table 4.

From Table 4, it can be seen that the natural gas pipeline reached the maximum moisture value of 45 at 4 pm, and the telecommunications pipe and water pipe reached the maximum moisture value at 10 pm. The peak periods of the power pipe and thermal pipes are 12 and 14, respectively

(2) Time analysis diagram

Time analysis map is the most common and important method for the safety visualization of underground space. It can display the number of changes in data, work progress, etc. through charts. This allows managers to inspect data intuitively, grasp key information in time, and make decisions at any time. Because the data obtained by the time analysis diagram can help projects have a predictive effect on hidden dangers and dangerous sources during construction . The time analysis chart plays a vital role in monitoring the safety management of underground space, analyzing the hidden dangers, and counting the sources of danger. The following takes the air-conditioning system of the underground space as an example to analyze the operation of the air-conditioning system.

The change trend of supply air and return air temperature over time is made into an analysis diagram of supply air and return air temperature, as shown in Fig. 6.

The analysis chart of supply air and return air temperature changes can give early warning according to the set maximum control value and minimum control value. In addition, trend analysis can be performed. It uses exponential, logarithmic, linear and other methods to perform regression analysis on the monitoring value, and monitor its trend change with time.

In the process of investigating and dealing with hidden safety hazards in underground space, managers need to grasp the current

Table 4
Humidity time data table of integrated pipe gallery.

Time	0	2	4	6	8	10	12	14	16	18	20	22	24
Natural Gas Pipeline	28	29	22	31	35	39	36	33	45	38	35	34	21
Telecom pipeline	40	42	40	38	44	39	34	48	43	36	50	52	45
water pipe	44	57	53	59	50	57	55	46	39	44	57	53	48
power pipes	23	21	35	33	39	34	41	37	28	33	37	32	26
heat pipe	15	13	17	20	19	15	22	28	25	19	18	19	13

management status of hidden dangers. It includes data entry status, hidden danger confirmation status, rectification and repair status, review prevention status, and punishment warning status [20]. According to different state hidden dangers, it comprehensively and systematically monitors underground security hidden dangers. As shown in Table 5.

4.2. Numerical simulation of underground space construction process

(1) Analysis of the initial state of the surface

In conventional structural design, when calculating the internal force of the superstructure and the additional settlement of the foundation, the superstructure and the foundation are calculated as two independent components. First of all, it uses the building foundation as a fixed support, and calculates the internal force of the structure and the reaction force of the foundation under the load of the superstructure. Then, the foundation reaction force is reacted to the elastic foundation soil, and then the foundation internal force and additional foundation settlement are calculated.Such as daily earthquakes, the interaction of soil and structure during earthquakes is a common problem However, this method does not take into account the effects of soil-structure interaction. As a result, some parts of the superstructure are unsafe. The simulation based on the digital twin can restore a real stress state of the earth's surface. While there are buildings on the surface, it can also simulate the actual force characteristics and underground structures.

Assuming that there is a building on the surface, it uses the digital twin model to simulate the building in the initial state, including the following steps:

Earth stress balance - existing building generation - building load application. In the initial state, the settlement change of the column bottom of the surface building is shown in Fig. 7:

It can be seen from Fig. 7 that after the construction of the underground space, the maximum settlement value of the surface building is 9.673 mm, which is in line with the settlement requirement of 3 cm in the current code. At the same time, it can be obtained that the maximum value of the settlement difference between the adjacent column foundations between the longitudinal side column X_3 and the corner column X_1 of the superstructure is 1.496 mm, which meets the requirements of the current code for the settlement difference of 1 cm between adjacent column foundations.

(2) Analysis of construction settlement law

The engineering construction mainly includes basement soil excavation and column side soil excavation. There are three main parts. The simulated construction process based on digital twin is shown in Fig. 8:

The settlement change of the column bottom of the superstructure during the construction process is shown in Fig. 9:

It can be seen from the data in the figure that during the construction stage, the settlement of the column bottom of the superstructure changed greatly. The change of the settlement difference between the adjacent column foundations of the superstructure is relatively gentle. When excavating large surface soil, the settlement of foundation is 5.6 mm. When the pile side soil is removed, the settlement of the surface building is larger, which is 0.8 mm. When the construction of the soil body on the side of the pile is completed, it indicates that the underpinning technology of the pile foundation of the surface building has been basically completed. The subsequent soil removal on the pile side has little effect on the superstructure. The final settlement difference at this stage is 2.427 mm. It complies with the requirement of 1 cm adjacent column foundation settlement difference stipulated by the current specification.

(3) Analysis of the difference between the settlement of the bottom of the building column and the settlement of the adjacent column foundation

During the construction of the underground space, the position of the central column is the maximum settlement of the column bottom of the superstructure. The maximum settlement difference between adjacent column foundations is between the longitudinal side column and the corner column of the surface building. From this, the maximum settlement of the column bottom and the

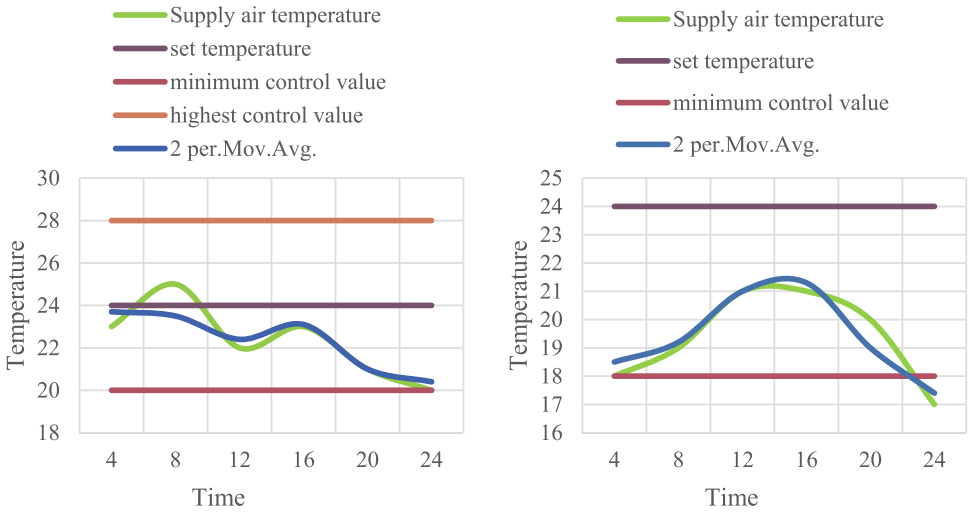


Fig. 6. Analysis of temperature changes of supply air and return air.

Table 5
Security hazard processing process diagram.

Hazard number	Security risks	States				
		Input data	Identify hidden dangers	Rectification and repair	Review and Prevention	Punishment and warning
#1	Lighting ballast burned out	√	√			
#2	The fire passage is blocked	√	√	√	√	√
#3	The camera at the north gate of the shopping mall is damaged	√	√	√		
#4	The fire extinguisher is too old	√	√	√	√	√
#5	The rolling shutter door of the South Gate falls off	√	√	√	√	

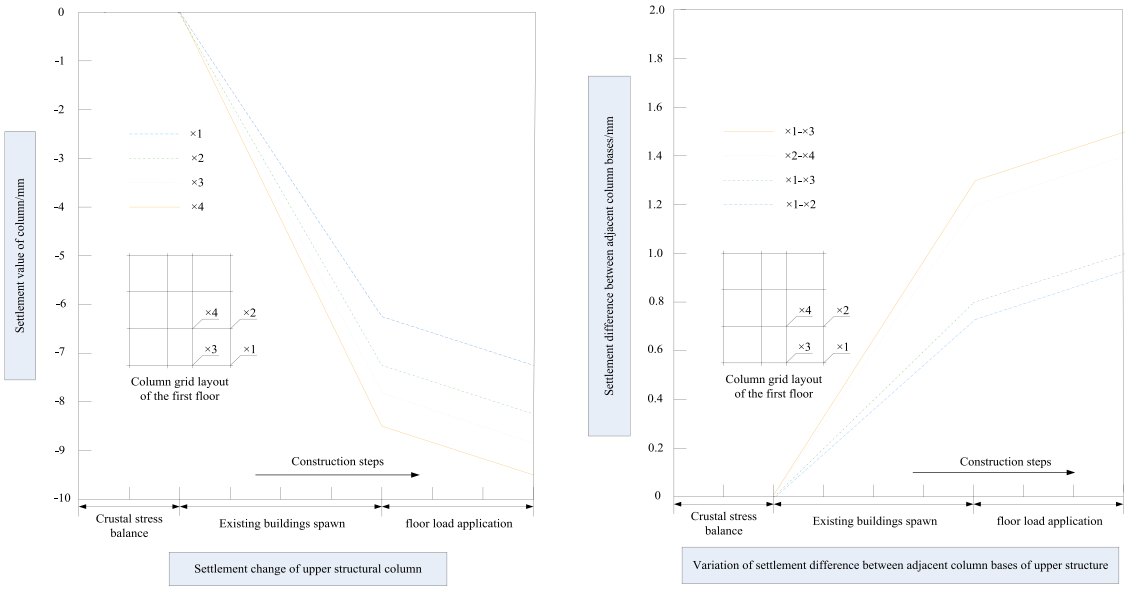


Fig. 7. The influence of construction on the settlement of the column bottom of the surface building in the initial state.

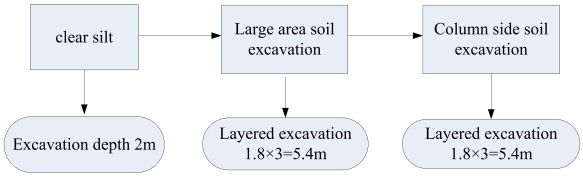


Fig. 8. Simulated construction process.

maximum settlement difference of the adjacent column bases in each stage can be obtained, as shown in Table 6:

The results show that there is friction between the pile body and the foundation soil during the construction process. It also penetrated deep into the ground, causing additional subsidence of the superstructure. It also has a "basin" settlement trend, which changes the additional internal force of the superstructure. As shown in Table 6, the maximum settlement of the column bottom is 21.897 mm, and the maximum settlement difference of adjacent column bases is 2.886 mm. It complies with the requirements of the current code for the settlement of the bottom of the building column 3 cm and the difference between the settlement of the adjacent column base 1 cm. Therefore, the underground space construction scheme based on the digital twin model is safe and compliant. If there are buildings on the surface, the normal use of the buildings on the surface can also be guaranteed.

4.3. Underground space fire detection and emergency countermeasures

The underground space is characterized by being closed and crowded. If a safety accident occurs, it will cause huge casualties and adverse social impact. Therefore, real-time monitoring of disaster prevention and timely rescue of disasters are the keys to ensuring the

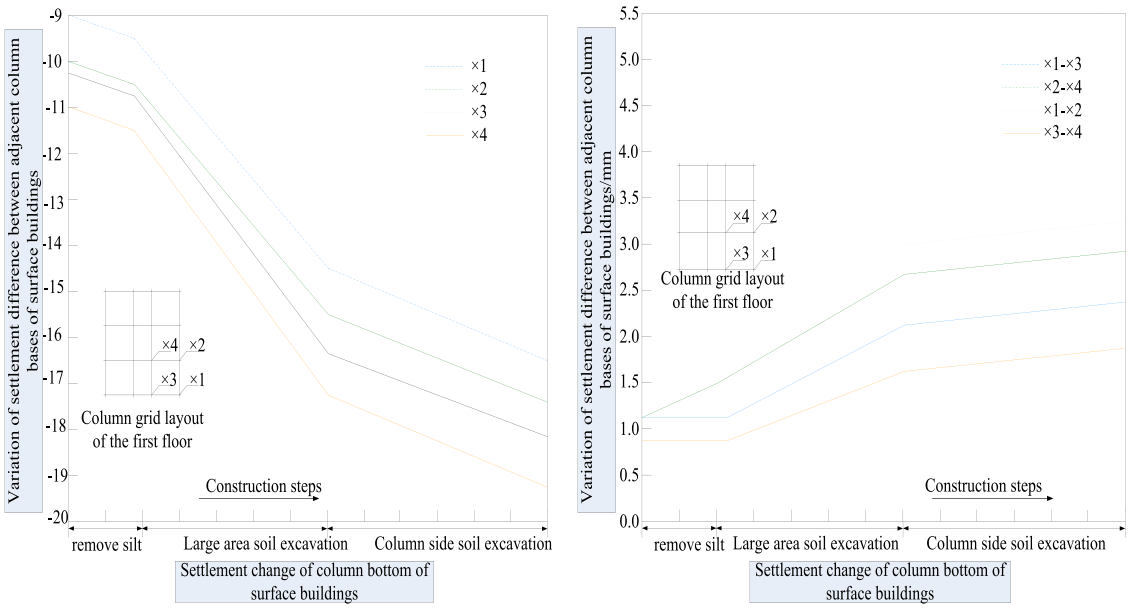


Fig. 9. The influence of the settlement value of the column bottom of the surface building.

Table 6
Influence of construction at each stage on settlement of superstructure (unit/mm).

Construction stage	Before underpinning	Underpinning construction		After underpinning
	Initial stage	Support system construction	earth excavation	Basement structure construction
Maximum settlement of middle column	−9.763	−12.247	−20.138	−21.897
Maximum settlement difference between adjacent column bases	1.510	1.207	2.283	2.886

safety and stability of underground space. A digital twin-based security monitoring system can effectively monitor the temperature changes of objects in underground space. It makes timely alarms and analysis of some extreme data to reduce the probability of disasters or minimize the losses caused by disasters.

When a fire occurs in the underground space, the multi -spectrum imaging instrument in digital twin technology can help detect and identify specific outline portraits, which can pass through the heat barrier caused by the fire scene Essence . When a fire breaks out in the underground space, the digital twin technology can penetrate the thermal barrier caused by the fire scene and clearly image key targets. The following is a comparison experiment of the imaging clarity of the detection images of the fire scene simulation for three spectrum imagers of 0.3–0.72 μm, 0.5–1.2 μm, and 8–16 μm. The results are shown in Fig. 10. The sharpness of the image is represented by ACM.

The experimental results show that no matter whether the experimental object is luminous or not, the size of smoke particles and the concentration of smoke will not affect the observation effect of infrared imaging. The temperature comparison between the experimental object and the surrounding environment is the only influencing factor. In practice, high temperature and high concentration of flue gas will make the environment more complicated. Under the high concentration of smoke, only the observation effect of infrared imaging can reach the ideal level. Neither shortwave visible light imaging nor normal visible light imaging will work. When the fire continued to expand, the underground environment was continuously affected by the thermal effect of high-temperature flue gas, and the temperature contrast was low. At this time, only the observation effect of short-wave infrared imaging can reach the ideal level. Neither infrared imaging nor ordinary visible light do anything. It is concluded that the monitoring system based on the fusion of short-wave infrared imaging and long-wave infrared imaging equipment can be applied to disaster prevention in underground space. The digital twin technology can better achieve this effect. And it is suitable for more complicated fire fighting actual fire environment.

From the above experiments, it can be concluded that the detection system based on digital twin technology can perform accurate testing in places with higher smoke concentrations.For some densely populated places such as subway stations, a blower can be added at the stairway to form airflow at the stairway and evacuate downward. This helps to slow down the speed of the smoke spreading to the upper layers. It can better improve the situation of casualties caused by the spread of smoke in the underground space.

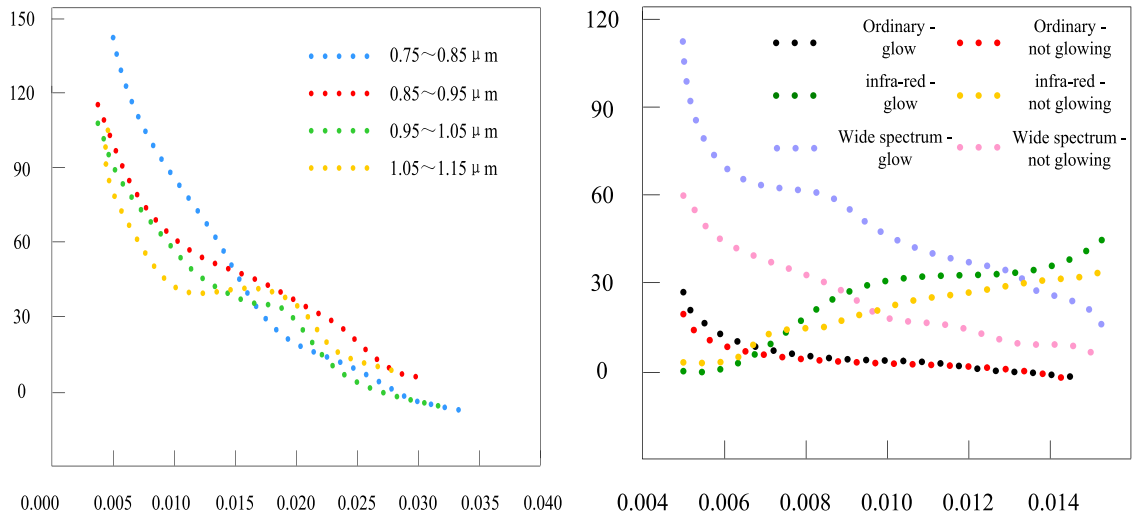


Fig. 10. The relationship between different smoke concentrations and image sharpness and the contrast of sharpness detected by the three spectrum imagers in different smoke concentrations.

5. Discussion

This article uses digital twin technology to design research on underground space . This paper is devoted to studying the optimization model based on digital twin and applying it to the intelligent overall planning of underground space. Because of the insufficient equipment and data collection before, the intelligent planning of the underground space was delayed. Provide some references for overall underground space planning on the basis of traditional planning. It not only expands the application scope of the digital twin model, but also is a new attempt for the overall planning of underground space. Through the simulation of underground space construction and fire accidents, the digital twin model is mined as an important method to study underground space. In order to optimize the overall planning of underground space and realize intelligent management, it needs to use digital twin technology as a guide to analyze each part of the construction and development of underground space in detail.

Through the analysis of this paper, it is shown that the application of digital twin technology to the intelligent overall planning of underground space is very effective. Digital twin technology can play an important role in the construction, monitoring and disaster prevention of underground space. This article first explains the digital twin model and introduces the five elements of the digital twin model. Then it briefly describes the algorithm of the digital twin, which evaluates the risk of safety accidents in underground space. It determines the methods of underground space development and construction and disaster prevention monitoring by simulating the construction process and safety accidents. The visualization data of underground space, the law of construction settlement and the most suitable imaging equipment for disaster prevention monitoring are obtained.

6. Conclusions

This article uses digital twin technology to design research on underground space . Through the analysis of this paper, the following conclusions are drawn: (1) The informatization and visualization method based on digital twin is an important means of underground space planning and management. Due to the characteristics of closed space, large negative externality, and difficulty in rescue, the traditional methods have been unable to meet the modern planning and management of underground space. The visualization technology under the guidance of digital twin technology can provide a strong guarantee for the overall planning of underground space. (2) Digital twin technology realizes digital collaboration through attribute information and associated information. It turns the underground space into a visual space that can be fully perceived. This is conducive to the construction of a new intelligent overall planning model for underground space. (3) The digital twin technology can monitor the underground space in real time, and monitor and analyze the real-time data. It will report abnormal data in time and correct errors. This helps to provide managers with more accurate, efficient and convenient management and decision-making. This provides greater possibilities for data judgment and analysis, the display of planning results, and the prevention and control of safety accidents. (4) During the construction of the underground space, based on the digital twin technology, the maximum settlement of the column bottom is 21.897 mm. The maximum settlement difference between adjacent column foundations is 2.886 mm. This is in line with the requirements of the current code for the settlement of the building column bottom of 3 cm and the difference between the settlement of the adjacent column base of 1 cm. When excavating a large area of soil, the settlement of the foundation is 5.6 mm. When the pile side soil is removed, the settlement of the surface building is 0.8 mm. This is in line with the 1 cm adjacent column foundation settlement difference requirement stipulated by the current code. This shows that the construction scheme based on digital twin technology is feasible, safe and standardized, and the impact on the surface buildings is also very small.

Funding

No funding was used to support this study.

Availability of data and materials

No data were used to support this study.

Declaration of competing interest

The authors declare that they have no competing interests.

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